(ii) Equation of circle with centre (*h, k*) and radius *r*

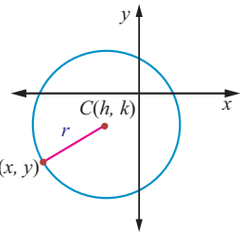


Fig.5.8

Let the centre be $C(h,k)$ and r be the radius and $P(x,y)$ be the moving point.

Then, $CP = r$ and so $CP^2 = r^2$.

That is, $(x-h)^2+(y-k)^2 = r^2$. This is the equation of the circle in Standard form, which is also known as centre-radius form.

Expanding the equation, we get

$x^2+y^2-2hx-2ky+h^2+k^2-r^2=0$.

Taking $2g=-2h, 2f=-2k, c=h^2+k^2-r^2$, the equation takes the form

$x^2+y^2+2gx+2fy+c=0$, called the general form of a circle.

The equation $x^2+y^2+2gx+2fy+c=0$ is a second degree equation in x and y possessing the following characteristics:

(i) It is a second degree equation in x and y,

(ii) coefficient of x^2 = coefficient of $y^2 \neq 0$,

(iii) coefficient of $xy = 0$.

Conversely, we prove that an equation possessing these three characteristics, always represents a circle. Let

$$ax^2+ay^2+2g'x+2f'y+c=0 \quad \ldots (1)$$

be a second degree equation in x and y having characteristics (i), (ii) and (iii), $a \neq 0$. Dividing (1) by a, gives

$$x^2+y^2+\frac{2g'}{a}x+\frac{2f'}{a}y+\frac{c'}{a}=0. \quad \ldots (2)$$

Taking $\frac{g'}{a}=g, \frac{f'}{a}=f$ and $\frac{c'}{a}=c$, equation (2) becomes $x^2+y^2+2gx+2fy+c=0$.

Adding and subtracting g^2 and f^2, we get $x^2+2gx+g^2+y^2+2fy+f^2-g^2-f^2+c=0$

$\Rightarrow (x+g)^2+(y+f)^2=g^2+f^2-c$

$\Rightarrow (x-(-g))^2+(y-(-f))^2=\left(\sqrt{g^2+f^2-c}\right)^2$

This is in the standard form of a circle with centre C $(-g,-f)$ and radius $r=\sqrt{g^2+f^2-c}$. Hence equation (1) represents a circle with centre $(-g,-f)=\left(\frac{-g'}{a},\frac{-f'}{a}\right)$ and radius

$=\sqrt{g^2+f^2-c}=\frac{1}{a}\sqrt{g'^2+f'^2-c'a}$.

Note

The equation of the circle $x^2+y^2+2gx+2fy+c=0$ with centre $(-g,-f)$ and radius $\sqrt{g^2+f^2-c}$ represents.

(i) a real circle if $g^2+f^2-c>0$;

(ii) a point circle if $g^2+f^2-c=0$;

(iii) an imaginary circle if $g^2+f^2-c<0$ with no locus.

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